Theorem: Let G be a graph and $c: E(G) \to \mathbb{N}_{\geq 1}$. Let $s, t \in V(G)$ and $s \neq t$. There exists a maximum (c,s,t)-flow $f: V(G) \times V(G) \to \mathbb{R}_{\geq 0}$ st $f(x,y) \in \mathbb{N}_{\geq 0}$ $\forall\, x,y \in V(G)$.

proof: **Recall**: $\min\{c(S,T) : (S,T) \text{ is an } s\text{-}t \text{ cut}\} \geq |f| = \sum_{x \in V(G)} f(s,x) - f(x,s)$ $\forall\, f$ (c,s,t)-flow.

Goal: Find a (c,s,t)-flow $f: V(G) \times V(G) \to \mathbb{N}_{\geq 0}$ and a s-t cut (S,T) such that $|f| \geq c(S,T)$.

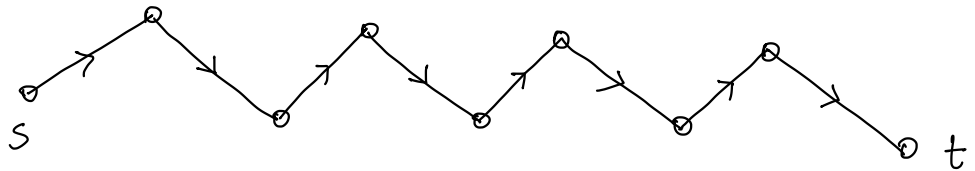
Let $i \geq 0$ and $f_i: V(G) \times V(G) \to \mathbb{N}_{\geq 0}$ be a flow (not necessarily maximum)

Let H_i with $V(H_i) = V(G)$ and $E(H_i) = \{(u,v) : f_i(u,v) < c(u,v) \text{ or } f_i(v,u) > 0\}$

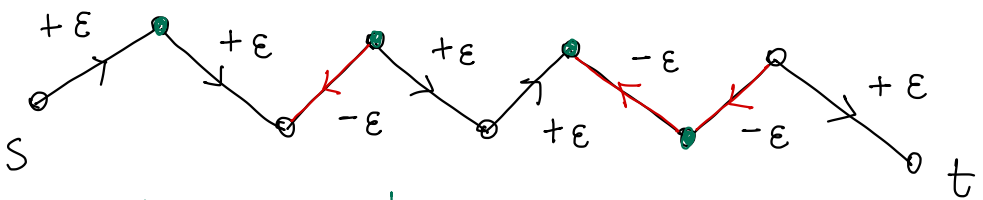
$S_i := \{v \in V(H) : \exists\, s \to v \text{ path in } H_i\}$

Case 1: $t \in S_i$

Let P be an $s \to t$ path in H_i.



H_i



G ← New flow: f_{i+1}

All configurations a vtx can see.

$$\varepsilon := \min\{c(u,v) - f(u,v) : (u,v) \in E(P),\ f(u,v) < c(u,v)\} \wedge \min\{f(v,u) : (u,v) \in E(P),\ f(v,u) > 0\}$$

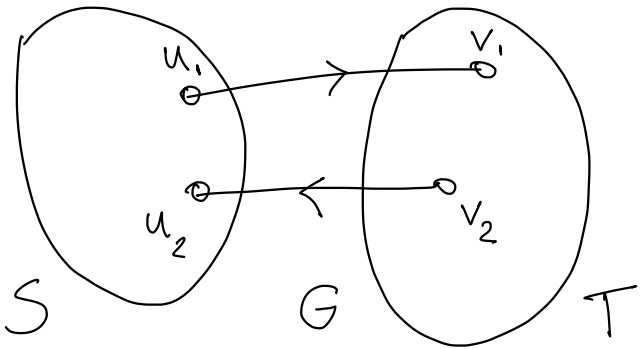
$$\in \mathbb{N}_{\geq 1}$$

Repeat until $t \notin S_i$ for some i. ∎

Case 2: $t \notin S_i$

Set $f = f_i$, $S = S_i$ and $T = V(G) \setminus S$. Note that (S,T) is an s-t cut.

Defn of S $\Rightarrow$



- $f(u_1,v_1) = c(u_1,v_1)$
- $f(u_2,v_2) = 0$.

$$\sum_{\substack{u \in S \cancel{\setminus \{s\}} \\ v \in V(G)}} f(u,v) - f(v,u) + \cancel{|f|} - |f| = 0$$

S

$$|f| = \sum_{\substack{u \in S \\ v \in V(G)}} f(u,v) - f(v,u) = \sum_{\substack{u \in S \\ v \in T}} f(u,v) - f(v,u) = \sum_{\substack{u \in S \\ v \in T}} c(u,v)$$

$$= C(S,T)$$

$\blacksquare$

Planar graphs

<u>Defn</u>: G is planar if $\exists$ injective map $p : V(G) \to \mathbb{R}^2$ and continuous maps $h_{uv} : [0,1] \to \mathbb{R}^2$ $\forall$ $uv \in E(G)$ such that

① $\{h_{uv}(0), h_{uv}(1)\} = \{p(u), p(v)\}$

② $\forall$ $uv, xy \in E(G)$ (distinct) we have

$$h_{xy}([0,1]) \cap h_{uv}([0,1]) = \{p(x), p(y)\} \cap \{p(u), p(v)\}.$$

Set $\mathbf{h} = \{h_{xy} : xy \in E(G)\}$. We refer to $(p, \mathbf{h})$ as a plane embedding of G.

$(G, p, \mathbf{h})$ = plane graph.

$\mathbf{h}(G) := \bigcup_{xy \in E(G)} h_{xy}([0,1])$ is a drawing of G. <u>Obs</u>: $\mathbf{h}(G)$ is a compact set.

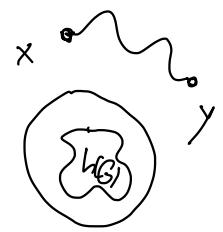
<u>Fáry's thm</u>: G is planar $\iff \exists$ plane embedding $\{h_{xy} : xy \in E(G)\}$ of G such that $h_{xy}([0,1])$ is a line segment in $\mathbb{R}^2$ $\forall\, xy \in E(G)$.

<u>Defn</u>: Let G be a planar graph and $(p, \mathbf{h})$ be a plane embedding of G. For $x \in \mathbb{R}^2$, define

$$F^{(x)} = \left\{ y \in \mathbb{R}^2 : \begin{array}{l} \exists\, g : [0,1] \to \mathbb{R}^2 \text{ continuous and injective st } g([0,1]) \subseteq \mathbb{R}^2 \setminus \mathbf{h}(G), \\ g(0) = x \text{ and } g(1) = y \end{array} \right\}$$

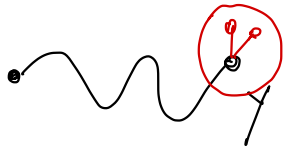
F is a <u>face</u> of (G, p, h) if $F = F^{(x)}$ for some x

$F(G)$ = collection of all distinct faces of G, $f(G) = \#$ faces



<u>Obs</u>: ① Every face is an open set.

② $h(G)$ is compact $\Rightarrow$ G has only one unbounded face. This is called <u>the outer face</u>. The other faces are the <u>inner faces</u>.

proof of ① : x  y $\quad x, y \in f^{(x)}$

in the open set $\mathbb{R}^2 \setminus \mathbf{h}(G)$

proof of ② : Suppose $\mathbf{h}(G) \subseteq B(0,R)$. Then, $\forall\, x, y \in B(0,R)^c$ $\exists$ continuous path in $B(0,R)^c$ connecting x to y.

<u>Defn</u>: We say that $X \subseteq \mathbb{R}^2$ is <u>path-connected</u> if $\forall\, p, q \in X$ $\exists$ continuous map $\varphi : [0,1] \to X$ such that $\varphi(0) = p$ and $\varphi(1) = q$.

<u>Defn</u>: An open set $U \subseteq \mathbb{R}^2$ is connected if $\forall\, \mathcal{O}_1, \mathcal{O}_2 \subseteq U$ open st $\mathcal{O}_1 \cap \mathcal{O}_2 = \phi$, we have $U \neq \mathcal{O}_1 \cup \mathcal{O}_2$.

<u>Thm</u>: Let $U \subseteq \mathbb{R}^2$ be an open set. U is connected $\iff$ U is path-connected.

<u>Thm</u> + <u>Obs</u> $\Rightarrow$ every face is a connected open set.

<u>Defn</u>: $p \in \mathbb{R}^2$ is <u>accessible from U</u> if $\exists$ continuous map $\varphi : [0,1] \to \mathbb{R}^2$ with $\varphi([0,1)) \subseteq U$ and $\varphi(1) = p$.

<u>Defn</u>: Let $\varphi : [0,1] \to \mathbb{R}^2$ be a continuous map such that $\varphi(0) = \varphi(1)$ and $\varphi|_{[0,1)}$ is injective. We say that $\varphi([0,1])$ is a Jordan curve.

<u>Defn</u>: The frontier (or boundary) of a set $S \subseteq \mathbb{R}^2$ is

$$\partial S = \{ x \in \mathbb{R}^2 : \forall r > 0,\ B(x,r) \cap S \neq \emptyset \text{ and } B(x,r) \cap (\mathbb{R}^2 \setminus S) \neq \emptyset \}$$

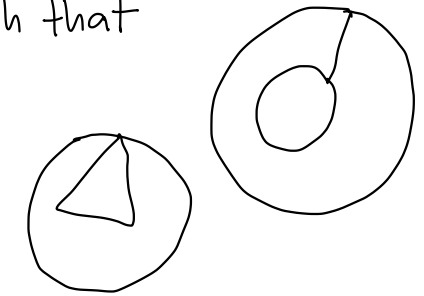
<u>Jordan's curve thm</u>: C Jordan curve in $\mathbb{R}^2 \Rightarrow \mathbb{R}^2 \setminus C = U_1 \cup U_2$, where U_1 and U_2 are connected disjoint open sets st $\partial U_1 = \partial U_2 = C$. Moreover, every $p \in C$ is accessible from U_1 and from U_2.

<u>Corollary</u>: Let F be a face of (G, p, h). Then,

$$\partial F = \{ p \in \partial F : p \text{ is acessible from } F \}.$$

Lemma: Let F be a face of (G, p, h). Then, $\exists\, H \subseteq G$ such that

$$\partial F = \bigcup_{xy \in E(H)} h_{xy}([0,1]).$$



Notation: $H = G[F]$.

Lemma: G forest $\Leftrightarrow$ G has only one face.

Lemma: # of faces F st $e \in E(G[F]) = \begin{cases} 1 & \text{if } \nexists \text{ cycle } C \text{ in } G \text{ st } e \in E(C). \\ 2 & \text{otherwise.} \end{cases}$

Lemma: G 2-connected planar, F face of $G \Rightarrow G[F]$ is a cycle.

For a proof, see Section 4.2 in Diestel's book.